

B2C Extension — Proposal

Project: RECAI Product Recommendation **Author:** Mounika Badri **Branch:** feature/b2c **Status:** For review

1. What we have today

The current RECAI application is an **internal dashboard for retail planners** — the people inside our business who decide which products should be pushed to which stores. When a planner logs in, they see:

- A **home dashboard** with high-level numbers (sales, top products, etc.).
- A **Store Recommendations** page that suggests which products each store should stock.
- A **Clusters** page that groups stores with similar buying patterns.
- An **Analysis / Forecast** page for future sales predictions.
- A **Chat Assistant** they can ask questions like *"which stores are in cluster 2?"*.

Everything the app knows about is at the **store level**. It answers questions like *"what should store S01001 stock next month?"* — but it has no idea who the individual shoppers walking into that store are.

2. What the senior asked for

Add a **B2C angle** — meaning, bring the **end customer / shopper** into the picture — but *without* rebuilding the existing app. Keep everything the planners already use, and add new capabilities alongside.

3. What "B2C" means, in two flavors

There are two very different ways to add "B2C" to the app. They lead to very different amounts of work.

Flavor	In simple words	What it needs
A. Customer insights for the planner	Give the planner a new set of pages that answer <i>"who are our customers and how do they behave?"</i> — but only planners see it. No customer ever logs in.	Just customer-behavior data. Ready to build now.
B. A real shopping experience for customers	Actual end-shoppers log in to the app, browse products, get personalized recommendations, place orders.	A real product catalog (names, images, prices), a proper backend login system, order/cart features. Significantly bigger effort.

Recommendation: do Flavor A first. It's fast, it reuses almost everything we already built, and it delivers immediate value to the planner. Flavor B stays on the roadmap but is *not* part of this proposal.

4. The data we'll use

I've already downloaded a public dataset called **UserBehavior** (from Alibaba's Taobao platform, around 100 million rows). For every shopper it records:

- Which product they **viewed**,

- Which product they **added to cart**,
- Which product they **favourited / wishlisted**,
- Which product they **bought**,
- And when each of these happened.

Honest limitation: this dataset has no product names, images, or prices — only anonymised IDs. That is perfectly fine for *analysing customer behaviour* (Flavor A), but it is **not** enough to run a real online store (Flavor B). If Flavor B is ever approved, we'd bring in a second dataset (Amazon product reviews) that has full product details.

5. What we'll actually add to the app (Flavor A)

Four new areas of value for the planner. Each translates into new pages or panels in the existing app — no separate app, no separate login.

5.1 A "Customers" section in the sidebar

Alongside the existing "Stores" area, a new "Customers" area with three pages:

1. **Customer list & search** — planner can search for any shopper and open their profile.
2. **Customer 360 profile** — for a chosen shopper, we show:
 - What categories they browse and buy the most.
 - How often they come back.
 - Where they usually drop off (e.g. they view a lot but rarely buy).
 - Which "type of customer" they are (see next point).
3. **Customer Segments** — we group all shoppers into a handful of easy-to-understand types, for example:
 - *"Frequent buyers"* — come back often, buy often.
 - "Browsers who rarely convert"* — look a lot, buy little.
 - "Big-basket occasional shoppers"* — visit rarely, but buy a lot when they do.
 - "New / one-time shoppers"* — bought once, haven't returned.

Each segment shows how many people are in it and what they tend to buy.

5.2 A "Shopping funnel" view

A simple visualisation showing, for any time period or product category, the customer drop-off:

View → Add to cart → Add to wishlist → Buy.

This tells the planner things like *"in Kids Outerwear, 80% of people who add to cart don't actually buy — we might have a pricing or checkout issue."*

5.3 A "Frequently bought together" panel

On the existing Store Recommendations page, we add a small extra tab that shows product pairs commonly bought together (like the *"Customers also bought"* strip on Amazon). This turns individual recommendations into **bundle suggestions**.

5.4 A smarter Chat Assistant

The existing chat assistant already answers store and cluster questions. We'll teach it a few new tricks so the planner can ask things like:

- "Tell me about customer 12345."
- "What are the top items for the frequent-buyers segment?"
- "What does the shopping funnel look like for Kids Outerwear?"

6. Does it need a separate customer login?

No. For everything in this proposal, the planner is the only user, so we keep the current single login exactly as it is. Nothing on the login screen changes.

If Flavor B is approved in the future, we would then add a "role" concept to the same login (planner vs. customer) — but that is a future decision, not part of this proposal.

7. How the app will look after the changes

Nothing existing is removed or restructured. The change is purely **additive**:

- The **sidebar** gets grouped into two sections: *Stores (existing)* and *Customers (new)*.
- The **home dashboard** gains a small strip of new tiles (total customers, active this week, buy rate).
- The **Store Recommendations** page gets one extra tab ("*Frequently bought together*").
- The **Chat Assistant** gains a few new question types.
- Three brand-new pages appear under *Customers* (List, Profile, Segments).

For a planner logging in today and a planner logging in after the change, the experience of using the existing pages is **identical**. They simply have more places to explore.

8. Risks and how we handle them

Risk	In plain words	How we handle it
Dataset is very large	~100 million rows may be slow to work with.	Start with a smaller, representative sample first.
No product names in the data	Screens will show "item 2268318" instead of a real product name. Feels less polished in demos.	Present it as category-level insights; if we ever need real product names, we'll add the Amazon dataset later.
Risk to the existing B2B app	Any change could accidentally break something planners already rely on.	The new work is kept in separate files with a <code>b2c_</code> prefix and a separate run command. The existing pipeline is not modified.
Client-side login isn't secure enough for real customers	Fine for internal planners, not fine for the public.	Only matters if Flavor B is approved. At that point, we'd move authentication to the backend.

9. Ask

Approval to proceed with **Flavor A** (customer insights for the planner) on the `feature/b2c` branch. Flavor B (a customer-facing shopping experience) to be revisited once Flavor A is live and reviewed.